BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 102301

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. On Hydrolysis sucrose produces Glucose and.....

(A) Maltose
(B) Galactose
(C) Fructose
(D) Lactose

Q2. Deficiency of vitamin -A- causes.

(A) Scruvy
(B) Rickets
(C) Aging
(D) Night blindness

Q3. What is the last layer of atmosphere.

(A) Troposphere
(B) Thermosphere
(C) Stratesphare
(D) Mesosphere

Q4. All salts cause the water become hard except.

(A) $CaCl_2$
(B) $Ca(HCO_3)_2$
(C) $MgSO_4$
(D) NaCl

Q5. The impurity present in mineral is called_____.

(A) Metal
(B) Ore
(C) Gangue
(D) Salt.

Q6. All are air pollutants EXCEPT_____.

(A) NO_2
(B) N_2
(C) SO_2
(D) SO_3

Q7. When the value of "Kc" is very large it shows that.

(A) Reaction is at equilibrium.
(B) Equilibrium will never be achieved.
(C) Reaction will move in forward direction.
(D) Reaction will move in back word direction.

Q8. Which one of the following is weak base.

(A) NaOH
(B) $Al(OH)_3$
(C) KOH
(D) LiOH

Q9. Pyridine belongs to _____ Organic compound.

(A) Hetrocyclic
(B) Homocyclic
(C) Aromatic
(D) Alicyclic

Q10. $AlCl_3$ is an acid according to the concept of_____.

(A) Arrhenious
(B) Lowery & Bronsted
(C) Lewis
(D) Both A & B

Q11. The functional group of Aldehyde.

(A) -OH
(B) -CHO
(C) -CO-
(D) -COOH

Q12. Which one of the following is Substitution reaction.

(A) Halogenation of alkynes.
(B) Halogenation of alkenes
(C) Halogenation of alkanes
(D) Oxidation of alkene

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Write an equilibrium constant "Kc" and the unit of chemical equation $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$.

(ii) Differenciate between monoacides and poly acids with examples.

(iii) When a clear liquid is placed in a beaker , how can you identify it as an acid , base or neutral (water)?

(iv) Compare Aromatic compounds with hetrocyclic organic compounds.

(v) Why Alkanes are inert in nature?

(vi) Write any four physical properties of Monosaccharides.

(vii) Differenciate between primary and secondary air pollutants.

(viii) How chemistry helps to maintain clean swiming pool ?

(ix) Give four uses of urea.

(x) What is Homologous series . Give three characteristics of homologous series?

(xi) What is ozone formation Justify your answer with two steps of ozone formation?

SECTION "C"

Maximum Marks: 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Explain the Law of Mass Action with help of General reaction. $A + B \rightleftharpoons C + D$

(b) [3] Write the names of any three salts along with their correct chemical formulas.

QIV.

(a) [4] How Friedrich wohler proved vital Force theory wrong.Discuss with proper example.

(b) [3] Write a balanced equation for the complete combustion of each of the following . (a) Methane (b) Ethane (c) Ethyne

QV.

(a) [4] Write the effect of Acid rain on (a) Plants & Trees (b) Materials.

(b) [3] Write the names of three vitamins and their importance.

QVI.

(a) [4] Describe the two methods i.e. Hand picking and hydraulic washing for the concentration of ore.

(b) [3] Give the balance reaction of water with i) K ii) Ca iii) Fe